

TOWARDS GLOBAL GEOMETRY PRESERVATION IN TEXT EMBEDDING DISTILLATION

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ABSTRACT

A text embedding teacher stores knowledge in the geometry it induces over an entire corpus, but supervising all pairwise relations is infeasible. Standard relational knowledge distillation (KD) instead forms its supervision support from examples that happen to co-occur in a mini-batch, so relation exposure is independent of semantic relatedness: a related pair and an unrelated corpus pair are equally likely to be supervised. We argue that a corpus-level relational objective should separate two decisions that mini-batch KD conflates—which relations are *eligible* for supervision, and how eligible relations are *realized* efficiently—and that preserving global geometry depends on both alignment, reproducing the supervised relations accurately, and coverage, supervising most of the relational mass the teacher considers important. We introduce **Global Geometry Preservation Knowledge Distillation (GGPKD)**, which freezes the base student’s k NN shortlists, asks the teacher to value the selected relations, and uses mini-batches only to realize them. GGPKD encodes the corpus once with the teacher and base student, then forms teacher-valued transition targets on the student-selected columns. The deduplicated candidate pool of a batch is encoded once and reused twice more: to calibrate similarity levels across the batch, and to supervise additional graph-selected rows without any extra student encoding. Independent coefficients weight the ambient and graph relational groups directly, alongside the graph width and auxiliary-row coefficient. Across nine standard sentence-embedding benchmarks and three teacher–student settings, GGPKD obtains the best mean score in every setting, with the largest and most consistent gains on semantic textual similarity. GGPKD requires no task labels and no online teacher inference, and only the student encoder is used at deployment.

1 INTRODUCTION

Text embedding models have become a fundamental component of modern NLP systems, supporting applications such as semantic search, retrieval, clustering, semantic textual similarity, and classification (Gao et al., 2023; Muennighoff et al., 2023). While larger embedding models can achieve strong performance, their computational and memory requirements may limit practical deployment, motivating the development of compact models that retain high-quality representations (Zhang et al., 2025). While larger text embedding models often achieve stronger performance on benchmark such as MTEB (Muennighoff et al., 2023), their increased parameter counts and representation dimensionality can incur substantial computational, memory, and deployment costs.

Knowledge distillation (KD) offers a practical way to reduce this deployment burden by transferring knowledge from a high-capacity teacher to a more compact student (Hinton et al., 2015). Recent embedding distillation frameworks have demonstrated strong empirical performance through direct teacher–student representation alignment (Vujanic & Rückstieß, 2026). For text embedding models, however, relational structure is also important, as many downstream applications operate directly on the similarities, distances, and relative ordering induced among embedded texts (Reimers & Gurevych, 2019; Muennighoff et al., 2023). Pointwise representation alignment does not explicitly constrain these inter-text relationships and may therefore leave important aspects of the teacher’s embedding geometry uncontrolled. This has motivated relational and structure-preserving distillation methods that explicitly transfer teacher-induced similarities, distances, angles, and other geometric

relationships (Tung & Mori, 2019; Park et al., 2019; Kim et al., 2023; Zhang et al., 2024; Dao et al., 2026).

At the corpus level, relational distillation would ideally account for the geometry induced among all training texts. Directly realizing such an objective, however, requires a quadratic number of pairwise relations. Relational KD is therefore commonly instantiated within mini-batches, where only relations among co-occurring texts are available for supervision (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021). Prior work has identified this batch locality as a limitation: batch-wise objectives observe only a fraction of dataset-level relations, with relational coverage becoming increasingly sparse as the dataset grows (Qian et al., 2022; Jung et al., 2023). Full-kernel approximations and memory-augmented methods have consequently been explored to extend relational supervision beyond individual batches (Qian et al., 2022; Jung et al., 2023; Yang et al., 2022b). These approaches broaden relational supervision beyond individual batches, but do not address which corpus relations should be prioritized under a limited relational budget. We formulate this as a support-selection problem: mini-batch KD lets batch co-occurrence decide which relations are supervised at all, whereas GGPKD freezes base-student retrieval support, lets the teacher value those relations, and uses mini-batches only to realize them.

We argue that this approximation introduces an under-emphasized bottleneck. Relational KD involves two distinct decisions: *which relations should be supervised and how those relations should be computed efficiently*. Mini-batch KD entangles the two. Although the relations themselves are defined by the teacher, their supervision support is determined by random batch co-occurrence. Consequently, an important teacher relation and an irrelevant corpus pair have the same probability of being exposed during training. Optimizing longer over such batches does not change which parts of the teacher geometry are observed, only how often the same uninformed sample is revisited. This observation suggests a different principle: *a frozen student retriever should determine which relations are eligible for supervision, the teacher should determine their targets, and mini-batches should only realize them efficiently*.

Based on this principle, we introduce GGPKD, a framework for student-selected, teacher-valued relational distillation. GGPKD encodes the corpus once with both the teacher and frozen base student. Student k NN selects each row’s columns; teacher cosines on those columns define the transition target, while the teacher’s own retrieval span defines its bandwidth. The supervision support is therefore fixed and identical in every epoch. Mini-batches enter only as a computational device: the candidate sets of a batch are deduplicated into a shared pool that is encoded once, and that single encoding pass supports three uses of the same vectors—matching each anchor’s transition row, calibrating similarity levels across the pool, and supervising the graph-selected rows of non-anchor pool members. Batch composition therefore determines which eligible relations can be realized in the current update, not which relations are considered relevant. Two explicit non-negative coefficients weight the relational groups directly, without normalization. Together with graph width and the auxiliary-row coefficient, this introduces four independent method-specific degrees of freedom.

Across nine datasets and three teacher–student settings, GGPKD attains the best mean score in all three settings. The gains are largest and most consistent on semantic textual similarity, while pair-classification results are less uniform. Our contributions are:

- We formulate relational knowledge distillation as a *support-selection* problem, and separate relation eligibility, which is fixed by frozen-student retrieval and valued by the teacher, from relation realization, which is a computational concern. Under this view, conventional mini-batch training selects relational supervision independently of teacher relevance, and corpus-level geometry preservation is controlled by two factors: alignment on the supervised relations and coverage of the teacher’s relational mass.
- We introduce GGPKD, which realizes this principle with a cached student- k NN, teacher-valued graph, per-row bandwidths derived from teacher retrieval spans, deterministic row-shaped supports, and amortized auxiliary-row supervision that reuses already-encoded pool members. The relational groups use explicit independent coefficients, giving four method-specific degrees of freedom including graph width and row weighting.
- We evaluate on nine datasets across three teacher–student settings, where GGPKD obtains the best mean score in every setting and the strongest results on semantic textual similarity,

and show that its explicit structural and loss settings are flat enough that the defaults are not selected on test results.

2 RELATED WORK

2.1 TEXT EMBEDDING DISTILLATION

Knowledge distillation transfers behavior from a large teacher to a compact student (Hinton et al., 2015). Language-model compression commonly matches output distributions, hidden states, or attention maps (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Xu et al., 2024), whereas embedding-specific methods target quantities used by similarity-based applications. Sparse Logit Sampling shows that deterministic top- K teacher logits induce truncation bias and uses randomized teacher-logit sampling to preserve the full teacher distribution in expectation (Anshumann et al., 2025). In that setting, however, the student still produces a full vocabulary vector; in corpus-level embedding KD, scoring an additional relation requires encoding another text, so support selection controls both approximation and encoder cost.

DistilCSE combines teacher supervision with contrastive learning (Gao et al., 2021; Xu et al., 2023); EmbedDistill transfers relative query–document geometry (Kim et al., 2023); and recent systems align embeddings, layers, tokens, or heterogeneous representation spaces (Lee et al., 2024; Zhang et al., 2024; Vujanic & Ruecksties, 2025; Truong et al., 2025; Dao et al., 2026; Akram et al., 2026). Relational approaches match pairwise similarities, distances and angles, conditional affinities, graph edges, or topological summaries (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021; Liu et al., 2019; Kim et al., 2024). Full Kernel Matrix Transfer uses Nystrom landmarks (Qian et al., 2022), and BINGO groups teacher-similar instances (Xu et al., 2022). Nevertheless, many relational objectives instantiate their relations from the current mini-batch; NRKD likewise mines an in-batch similarity matrix (Gou et al., 2025). This couples computational batching with geometric support selection.

2.2 BEYOND MINI-BATCH RELATIONAL SUPERVISION

Neighborhood graphs recover manifold structure in Laplacian Eigenmaps and Diffusion Maps, where a row-stochastic transition matrix over a k NN graph serves as the primitive description of local geometry (Belkin & Niyogi, 2003; Coifman & Lafon, 2006); heat-based constructions summarize the same operator at multiple scales (Sun et al., 2009; Huguet et al., 2023). Geometric KD aligns heat kernels between teacher and student graphs (Yang et al., 2022a), and graph-to-MLP methods distill observed graph structure into graphless models (Zhang et al., 2022; Tian et al., 2025); both assume a graph given by the task. GGPKD instead constructs the graph from cached text embeddings, so the teacher’s own retrieval behavior defines which relations a standard text encoder is supervised on. We use the transition operator at a single hop rather than its diffused powers: the one-hop rows already provide a teacher-weighted support and a per-row bandwidth, and multi-hop targets are treated as an alternative the method does not require. Neither the graph nor the teacher is needed at inference.

3 GGPKD: GLOBAL GEOMETRY THROUGH LOCAL ALIGNMENT

GGPKD distills an embedding teacher into a smaller student on neighborhoods retrieved once by the frozen base student. The student selects each row’s columns, while the teacher supplies the cosine targets over those columns. The method exposes graph width `graph_k`, relational coefficients `r0_weight/r1_weight`, and auxiliary-row weight `row_weight`. These three loss coefficients are independent and applied directly, without normalization. All temperatures and targets are derived from the cached embeddings. The graph and teacher embeddings are cached before training and are not needed at inference.

3.1 STUDENT-SELECTED, TEACHER-VALUED GRAPH

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be the deduplicated training corpus. The teacher and frozen base student each encode every text once, and their embeddings are ℓ_2 -normalized. For each node i , we retrieve the

$k = \text{graph_k}$ most similar non-self nodes under the base student, forming $\mathcal{N}_i^S$. If $s_i^{T,(q)}$ is the teacher’s q -th largest corpus cosine, the row bandwidth is

$$\tau_i = \frac{s_i^{T,(1)} - s_i^{T,(k)}}{\log k}. \quad (1)$$

On the teacher’s own reference top- k , the k -th neighbor receives $1/k$ of the nearest neighbor’s unnormalized weight. The target below is evaluated on the student-selected set and need not have that same endpoint ratio. Because the bandwidth scales with any affine rescaling of the similarities, the resulting softmax row is affine-invariant and requires no teacher-specific temperature tuning.

The neighbor set is the student’s retrieved list itself, $\mathcal{N}_i^S = \text{top-}k_S(i)$, and every node has degree exactly k . Teacher-kNN is retained as a source ablation. A mutual filter, which keeps $i \rightarrow j$ only when the two nodes retrieve each other, suppresses hubs but discards directed relations; we report it as an ablation in Section 4. The teacher transition row is

$$P_i^T(j) = \frac{\mathbf{1}[j \in \mathcal{N}_i^S] \exp(c_{ij}^T/\tau_i)}{\sum_{u \in \mathcal{N}_i^S} \exp(c_{iu}^T/\tau_i)}. \quad (2)$$

The bandwidth in Equation (1) is read from a teacher reference list of the same fixed width for every node; the target values are teacher cosines evaluated on the student-selected columns.

3.2 DETERMINISTIC CANDIDATE POOL

The candidate set for anchor i is exactly $\mathcal{N}_i^S$. Every column the base student retrieved is supervised and none is discarded, so no truncation constant enters the objective and every anchor is scored on the same number of columns, k . GGPkd draws no hard or random negatives and performs no per-epoch sampling: the candidates are a deterministic function of the graph and remain fixed throughout training. Within a mini-batch B , candidates are deduplicated into the shared pool

$$\mathcal{P}_B = \bigcup_{i \in B} \mathcal{N}_i^S, \quad (3)$$

so each distinct candidate text is encoded once.

3.3 TRAINING OBJECTIVE

The graph-scale loss matches each teacher row with a student softmax over the same columns and at the same stored bandwidth:

$$\mathcal{L}_{r=1} = \frac{1}{|B|} \sum_{i \in B} D_{\text{KL}} \left(P_i^T \parallel \text{softmax}_{j \in \mathcal{N}_i^S} (c_{ij}^S/\tau_i) \right). \quad (4)$$

This local row ranks candidates within an anchor’s own neighborhood. To also calibrate similarities across the batch, the ambient loss uses the complete shared pool. Its temperature is derived from the graph as $\tau_0 = \text{median}_i \tau_i$ and is shared by teacher and student:

$$\mathcal{L}_{r=0} = \frac{1}{|B|} \sum_{i \in B} D_{\text{KL}} \left(\text{softmax}_{j \in \mathcal{P}_B \setminus \{i\}} (c_{ij}^T/\tau_0) \parallel \text{softmax}_{j \in \mathcal{P}_B \setminus \{i\}} (c_{ij}^S/\tau_0) \right). \quad (5)$$

The ambient and graph groups receive explicit, independent non-negative coefficients λ_0 and λ_1 , applied directly.

GGPKD also reuses non-anchor candidates as auxiliary row centers. Let $\mathcal{J}_B = \mathcal{P}_B \setminus B$ and expose the portion of row j that is already present in the shared pool:

$$\Omega_j^B = \mathcal{N}_j^S \cap \mathcal{P}_B, \quad \mathcal{J}_B^+ = \{j \in \mathcal{J}_B : |\Omega_j^B| \geq 2\}. \quad (6)$$

Batch anchors are excluded because $\mathcal{L}_{r=1}$ already supervises their transition rows. The remaining rows are matched uniformly at their stored bandwidths:

$$\mathcal{L}_{\text{row}} = \frac{1}{|\mathcal{J}_B^+|} \sum_{j \in \mathcal{J}_B^+} D_{\text{KL}} \left(P_j^T|_{\Omega_j^B} \parallel \text{softmax}_{u \in \Omega_j^B} (c_{ju}^S/\tau_j) \right), \quad (7)$$

where the restricted teacher row is renormalized on Ω_j^B . This term requires no additional student encoding because all centers and columns are already in $\mathcal{P}_B$.

The complete objective is

$$\mathcal{L}_{\text{GGPKD}} = \underbrace{\lambda_0 \mathcal{L}_{r=0} + \lambda_1 \mathcal{L}_{r=1}}_{\mathcal{L}_{\text{rel}}} + \lambda_{\text{row}} \mathcal{L}_{\text{row}}. \quad (8)$$

The defaults are $\lambda_0 = \lambda_1 = 0.5$, `graph_k` = 200, and $\lambda_{\text{row}} = \text{row_weight} = 1$. No coefficient is divided by the sum of the others: common rescaling of λ_0 and λ_1 changes their total strength relative to λ_{row} . Increasing `graph_k` changes both neighborhood width and the bandwidth in Equation (1); those effects cannot be separated by a single-axis sweep.

Cost. Exact graph construction requires $O(N^2d)$ similarity work, although an approximate nearest-neighbor backend can replace the all-pairs search. During training, the dominant cost is encoding the deduplicated shared pool; $\mathcal{L}_{\text{row}}$ reuses those embeddings. At deployment, inference uses only the student encoder.

4 EXPERIMENTS

Table 1: Results across three teacher $\rightarrow$ student settings on nine datasets. Emotion, WiC, and STS-B (marked with $\star$) are in-domain; the remaining datasets are out-of-domain. Bold and underlined values denote the best and second-best results within each setting, excluding the Teacher and Student base reference rows. TALAS and GGPKD results are reported as mean $\pm$ standard deviation; the remaining baseline results are from TALAS (Dao et al., 2026). For setting (c), the epoch checkpoint is selected on the validation average.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Domain Avg		Avg.
	Banking77	Tweet	Emotion $\star$	MRPC	SciTail	WiC $\star$	SICK	STS12	STS-B $\star$	Avg-In	Avg-Out	Avg.
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	71.53	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	<u>73.83</u>	65.98	<u>83.33</u>	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	<u>90.94</u>	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	90.44 $\pm$ 0.13	74.62 $\pm$ 0.17	70.87 $\pm$ 0.52	85.90 $\pm$ 0.10	81.78 $\pm$ 0.29	68.84 $\pm$ 0.15	77.56 $\pm$ 0.10	73.12 $\pm$ 0.30	79.24 $\pm$ 0.10	72.98 $\pm$ 0.19	80.57 $\pm$ 0.01	78.04 $\pm$ 0.07
GGPKD	91.98 $\pm$ 0.09	73.74 $\pm$ 0.26	<u>68.78 $\pm$ 0.30</u>	84.22 $\pm$ 0.13	85.30 $\pm$ 0.16	66.28 $\pm$ 0.03	77.50 $\pm$ 0.06	75.95 $\pm$ 0.16	82.15 $\pm$ 0.07	72.40 $\pm$ 0.10	81.45 $\pm$ 0.12	78.43 $\pm$ 0.10
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	44.24	62.41	56.35
SimCSE-unsup	87.89	71.45	55.65	83.87	76.32	67.96	66.93	59.66	63.97	62.53	74.35	70.41
CDM	89.15	70.48	58.71	84.16	75.91	70.00	68.14	62.74	67.78	65.50	75.10	71.90
DSKD	89.51	70.53	57.27	84.39	75.36	<u>69.96</u>	68.81	66.78	70.05	65.76	75.90	72.52
Jasper and Stella	88.38	74.13	62.81	<u>85.98</u>	80.33	68.38	74.35	66.80	74.55	68.58	78.33	75.08
DistillCSE	90.50	72.70	59.88	84.39	78.48	69.49	73.97	70.00	73.73	67.70	78.34	74.79
EMO	90.19	<u>74.26</u>	61.48	85.31	80.82	68.30	75.41	67.06	76.35	68.71	78.84	75.46
TALAS	89.72 $\pm$ 0.19	75.38 $\pm$ 0.11	65.83 $\pm$ 0.58	87.58 $\pm$ 0.02	82.90 $\pm$ 0.12	66.96 $\pm$ 0.07	75.82 $\pm$ 0.11	68.11 $\pm$ 0.20	77.51 $\pm$ 0.18	70.10 $\pm$ 0.13	79.92 $\pm$ 0.05	76.64 $\pm$ 0.04
GGPKD	90.78 $\pm$ 0.09	73.57 $\pm$ 0.32	66.45 $\pm$ 0.20	85.13 $\pm$ 0.06	83.50 $\pm$ 0.20	62.75 $\pm$ 0.20	78.27 $\pm$ 0.06	74.94 $\pm$ 0.18	81.74 $\pm$ 0.07	70.31 $\pm$ 0.03	81.03 $\pm$ 0.08	77.46 $\pm$ 0.05
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	40.91	58.45	52.60
SimCSE-unsup	86.67	69.56	54.00	82.29	73.76	65.71	66.04	59.11	62.08	60.60	72.91	68.80
CDM	86.07	70.32	53.04	81.97	72.97	68.34	65.97	61.12	65.75	62.38	73.07	69.51
DSKD	85.66	69.86	52.25	82.17	73.52	<u>68.58</u>	66.54	62.52	66.70	62.51	73.38	69.76
Jasper and Stella	85.94	71.25	57.99	83.36	76.17	68.25	70.57	63.81	70.11	65.45	75.18	71.94
DistillCSE	88.03	69.87	54.24	83.20	77.66	67.88	70.09	68.19	71.53	64.55	76.17	72.30
EMO	87.03	<u>71.67</u>	59.63	83.45	78.17	68.89	<u>70.81</u>	65.78	71.42	<u>66.65</u>	76.15	72.98
TALAS	84.16 $\pm$ 0.15	72.82 $\pm$ 0.11	61.13 $\pm$ 0.52	84.85 $\pm$ 0.10	80.91 $\pm$ 0.27	66.32 $\pm$ 0.07	70.72 $\pm$ 0.15	67.80 $\pm$ 0.08	72.39 $\pm$ 0.31	66.61 $\pm$ 0.24	76.88 $\pm$ 0.07	73.46 $\pm$ 0.10
GGPKD	88.18 $\pm$ 0.03	70.78 $\pm$ 0.17	62.69 $\pm$ 0.14	<u>83.71 $\pm$ 0.07</u>	82.77 $\pm$ 0.11	65.39 $\pm$ 0.01	73.90 $\pm$ 0.08	74.57 $\pm$ 0.17	78.22 $\pm$ 0.02	68.77 $\pm$ 0.04	78.99 $\pm$ 0.05	75.58 $\pm$ 0.03

4.1 SENSITIVITY ANALYSIS

We evaluate GGPKD sensitivity on the Qwen3-Embedding-0.6B→MiniLMv2-H384 setting using seeds 42, 43, and 44. Tables 2 and 3 report downstream quality together with the measured resource cost. Increasing `graph_k` from 50 to 400 changes Avg. by only 0.34 points, but the largest setting costs substantially more time and GPU memory. Relative to the default `graph_k` = 200, the 400-neighbor setting gains 0.05 points while increasing mean training time by 36.8% and peak GPU memory by 54.2%. The row-loss sweep is also flat: Avg. spans 0.18 points, and the small timing differences are not sufficient to establish a causal cost effect. We therefore retain `graph_k` = 200 and $\lambda_{\text{row}} = 1$ as defaults rather than selecting a new setting on the test results.

5 CONCLUSION

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Table 2: Sensitivity to the student-graph neighborhood size for Qwen3-Embedding-0.6B→MiniLMv2-H384. Downstream scores and training wall time are mean $\pm$ sample standard deviation over seeds 42, 43, and 44. Training-memory columns report the maximum per-run peak across those seeds; graph-build measurements contain one build per setting. Host and GPU memory are reported in GiB.

graph_k	Downstream quality				Training cost			Graph-build cost		
	Avg-In	Avg-Out	Avg. $\uparrow$	Δ Avg.	Wall (s)	Host	GPU	Wall (s)	Host	GPU
50	68.29 $\pm$ 0.09	78.82 $\pm$ 0.03	75.31 $\pm$ 0.04	-0.29	231.53 $\pm$ 16.35	9.82	7.98	36.8	2.19	6.44
100	68.46 $\pm$ 0.22	79.02 $\pm$ 0.09	75.50 $\pm$ 0.07	-0.10	257.23 $\pm$ 35.31	9.80	9.89	20.9	1.56	0.96
200 (default)	68.79 $\pm$ 0.07	79.00 $\pm$ 0.03	75.60 $\pm$ 0.04	ref.	329.90 $\pm$ 8.68	10.15	11.50	21.8	1.63	0.96
400	68.89 $\pm$ 0.14	79.02 $\pm$ 0.00	75.64 $\pm$ 0.04	+0.05	451.30 $\pm$ 18.93	10.59	17.74	28.6	2.11	0.98

Table 3: Sensitivity to the auxiliary row-relation weight for Qwen3-Embedding-0.6B→MiniLMv2-H384 at the default graph_k = 200. Downstream scores and wall time are mean $\pm$ sample standard deviation over seeds 42, 43, and 44. Host and GPU columns report the maximum observed per-run peak in GiB.

λ_{row}	Downstream quality				Training cost		
	Avg-In	Avg-Out	Avg. $\uparrow$	Δ Avg.	Wall (s)	Host	GPU
0	68.63 $\pm$ 0.15	78.86 $\pm$ 0.03	75.45 $\pm$ 0.04	-0.15	326.00 $\pm$ 10.06	10.08	11.97
0.25	68.73 $\pm$ 0.19	78.91 $\pm$ 0.06	75.52 $\pm$ 0.07	-0.08	331.53 $\pm$ 10.78	10.15	11.50
0.5	68.64 $\pm$ 0.11	78.99 $\pm$ 0.10	75.54 $\pm$ 0.03	-0.06	350.43 $\pm$ 19.74	10.14	11.50
1.0 (default)	68.79 $\pm$ 0.07	79.00 $\pm$ 0.03	75.60 $\pm$ 0.04	ref.	329.90 $\pm$ 8.68	10.15	11.50
2.0	68.91 $\pm$ 0.14	78.99 $\pm$ 0.01	75.63 $\pm$ 0.05	+0.03	301.23 $\pm$ 33.81	10.17	11.50

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